

DIFFERENTIAL EQUATIONS

An equation in which differential coefficients occur is called a differential equation

Examples

$$(1) (2x+3y) \frac{dy}{dx} + (3x-4)y = 0$$

$$(2) \frac{d^2y}{dx^2} + y = 0$$

$$(3) \frac{d^2y}{dx^2} - \left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2} = 0$$

$$(4) \frac{\partial^2 z}{\partial x^2} + \frac{\partial^2 z}{\partial y^2} = 0$$

$$(5) \left(\frac{d^3y}{dx^3}\right)^2 - 3x\left(\frac{dy}{dx}\right)^5 + 5y = 0$$

$$(6) x \frac{\partial^2 z}{\partial x^2} - y \frac{\partial^2 z}{\partial y^2} = x^2 - y^2$$

Ordinary differential equations

A differential equation involving a single independent variable and hence ordinary derivatives is called an ordinary diff. eq

Examples (1), (2), (3) and (5) are ordinary diff. eqs.

Partial differential equations

If there are two or more independent variables present, so that the equation contains partial derivatives, it is called a partial diff. eq.

Examples (4) and (6) are partial diff. eqs.

Order of a diff. eq order of a diff. eq

is the order of the highest order derivative occurring in it.

Degree of a diff. eq.

degree of a diff. eq

is the degree of the highest order derivative occurring in it.

Example (2) first order first degree

eq (2) order 2 degree 1

eq (3) can be written as

$$\frac{d^2 y}{dx^2} = \left(1 + \left(\frac{dy}{dx}\right)^2\right)^{3/2} \text{ squaring } \left(\frac{d^2 y}{dx^2}\right)^2 = \left[1 + \left(\frac{dy}{dx}\right)^2\right]^3$$

Now fractional power is removed

$\therefore$ It is of order 2 and degree 2

eq (4) order-2 and degree 1 partial diff. eq

eq (5) order-3 degree 2

eq (6) is a first order first degree partial diff. eq.

Solution of a diff. eq

A relation between the variables not involving derivatives which satisfy the given diff. eq is called a solution of the diff. eq. A general solution of a diff. eq contains as many arbitrary constants as order of the eq. Giving particular values to arbitrary constants we get particular solutions.

Example consider the diff. eq. $\frac{dy}{dx} - x = 0$

$$\text{i.e. } \frac{dy}{dx} = x \Rightarrow dy = x dx$$

$$\text{Integrating } \int dy = \int x dx$$

$$\text{i.e. } y = \frac{x^2}{2} + \frac{c}{2} \Rightarrow 2y = x^2 + c$$

is the general solution

putting $c=0$, we get the particular soln $2y=x^2$, and putting $c=1$, we get the particular solution $2y=x^2+1$

Note consider the relation $2y=x^2+c$

diff. w.r. to x , $2 \frac{dy}{dx} = 2x \Rightarrow \frac{dy}{dx} = x$

i.e. $\frac{dy}{dx} - x = 0$ which is the diff. eq corresponding to the above relation

FIRST ORDER FIRST DEGREE EQUATIONS

Type I Variable separable equations

A diff. eq of the form $M(x,y)dx + N(x,y)dy = 0$ is a variable separable eq. if it can be expressed with the form $f(x)dx + g(y)dy = 0$

Solve (1) $\frac{dy}{dx} = \frac{y}{x}$ (2) $(y+2)dx + y(x+4)dy = 0$

$$(3) x \sin y dx + (x^2+1) \cos y dy = 0$$

$$(4) 4xy dx + (x^2+1) dy = 0$$

$$(5) \tan \theta dr + 2r d\theta = 0$$

$$(1) \frac{dy}{dx} = \frac{y}{x} \Rightarrow \frac{dy}{y} = \frac{dx}{x}$$

$$\text{Integrating, } \int \frac{dy}{y} = \int \frac{dx}{x} \Rightarrow \log y = \log x + \log c$$

$$\text{i.e. } \log y = \log(cx) \Rightarrow y = cx \text{ is the general}$$

solution.

$$(2) (y+2)dx + y(x+4)dy = 0$$

Dividing by: $(x+4)(y+2)$

$$\frac{dx}{x+4} + \frac{y}{y+2} dy = 0 \Rightarrow \frac{dx}{x+4} + \frac{y+2-2}{y+2} dy = 0$$

$$\text{i.e. } \frac{dx}{x+4} + \left(1 - \frac{2}{y+2}\right) dy = 0$$

Integrating $\log(x+4) + y - 2\log(y+2) = C$
is the general solution

$$(3) \quad x \sin y \, dx + (x^2+1) \cos y \, dy = 0$$

Dividing by $(x^2+1) \sin y$

$$\frac{x}{x^2+1} dx + \frac{\cos y}{\sin y} dy = 0$$

$$\therefore \int \frac{x}{x^2+1} dx + \int \frac{\cos y}{\sin y} dy = 0$$

$$\text{Multiplying by 2, } \int \frac{2x}{x^2+1} dx + 2 \int \frac{\cos y}{\sin y} dy = 0$$

$$\therefore \log(x^2+1) + 2\log \sin y = \log C$$

$$\therefore \log(x^2+1) \sin^2 y = \log C \Rightarrow (x^2+1) \sin^2 y = C$$

$$(4) \quad 4xy \, dx + (x^2+1) dy = 0$$

Dividing by $(x^2+1)y$,

$$\frac{4x \, dx}{x^2+1} + \frac{dy}{y} = 0$$

$$\text{Integrating } \int \frac{4x}{x^2+1} dx + \int \frac{dy}{y} = 0$$

$$\therefore 2 \int \frac{2x}{x^2+1} dx + \int \frac{dy}{y} = 0$$

$$\therefore 2\log(x^2+1) + \log y = \log C$$

$$\log(x^2+1)^2 y = \log C$$

soln is given by $(x^2+1)^2 y = C$

$$(5) \quad r \tan \theta \, dr + 2r \, d\theta$$

$$\text{Dividing } r \tan \theta, \quad \frac{dr}{r} + \frac{2}{\tan \theta} d\theta = 0$$

$$\Rightarrow \int \frac{dr}{r} + 2 \int \cot \theta \, d\theta = 0$$

$$\Rightarrow \log r + 2\log \sin \theta = \log C$$

$$\Rightarrow \log r \sin^2 \theta = \log C$$

$\therefore r \sin^2 \theta = C$ is the solution.

Homogeneous equations

A diff. eq which can be reduced to the form $\frac{dy}{dx} = F\left(\frac{y}{x}\right)$ is called a homogeneous diff. eq. This eq can be solved by putting $y = vx$ and hence reducing it to variable separable form

Solve the following diff. eqs

$$(1) \quad 2xy \frac{dy}{dx} - y^2 + x^2 = 0$$

$$(2) \quad xy' = x + y$$

$$(3) \quad \frac{dy}{dx} = \frac{y+x}{y-x}$$

$$(4) \quad (y + \sqrt{x^2 + y^2})dx - xdy = 0$$

$$(1) \quad 2xy \frac{dy}{dx} = y^2 - x^2 \Rightarrow \frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2 \left(\frac{y^2}{x^2} - 1 \right)}{2xy} \Rightarrow \frac{dy}{dx} = \frac{\left(\frac{y}{x} \right)^2 - 1}{2 \left(\frac{y}{x} \right)} \quad -xv$$

$\therefore$ It is a homogeneous diff. eq

$$\text{Put } y = vx \quad \therefore \frac{dy}{dx} = v + x \cdot \frac{dv}{dx}$$

$$\text{Sub in eq } v + x \frac{dv}{dx} = \frac{v^2 - 1}{2v}$$

$$\therefore x \frac{dv}{dx} = \frac{v^2 - 1}{2v} - v = \frac{v^2 - 1 - 2v^2}{2v}$$

$$\therefore x \frac{dv}{dx} = -\frac{(v^2 + 1)}{2v} \Rightarrow \frac{2v}{v^2 + 1} dv = -\frac{dx}{x}$$

$$\text{Integrating, } \int \frac{2v}{v^2 + 1} dv + \int \frac{dx}{x} = 0$$

$$\text{ie } \log(v^2 + 1) + \log x = \log c$$

$$\text{ie } \log(v^2 + 1)x = \log c$$

$$\text{ie } (v^2 + 1)x = c$$

$$\text{ie } \left(\frac{y^2}{x^2} + 1 \right) x = c$$

$$\text{ie } \frac{y^2 + x^2}{x^2} \cdot x = c \Rightarrow \underline{\underline{x^2 + y^2 = cx}}$$

$$(\because v = \frac{y}{x})$$

$$(2) \quad xy' = x + y \Rightarrow \frac{dy}{dx} = \frac{x+y}{x} \rightarrow (1)$$

$$\Rightarrow \frac{dy}{dx} = 1 + \frac{y}{x}$$

$$\text{Put } y = vx \quad \therefore \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\text{Eq (1) reduces to } v + x \frac{dv}{dx} = 1 + v$$

$$\text{or } x \frac{dv}{dx} = 1 \Rightarrow dv = \frac{dx}{x}$$

$$\text{Integrating } v = \log x + \log c$$

$$\text{or } v = \log cx \Rightarrow \frac{y}{x} = \log cx$$

$\therefore y = x \log cx$ is the solution.

$$(3) \quad \frac{dy}{dx} = \frac{y+x}{y-x} \Rightarrow \frac{dy}{dx} = \frac{x(\frac{y}{x}+1)}{x(\frac{y}{x}-1)} = \frac{\frac{y}{x}+1}{\frac{y}{x}-1}$$

$$\text{put } y = vx, \quad \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\therefore \text{ we get } v + x \frac{dv}{dx} = \frac{v+1}{v-1}$$

$$\therefore x \frac{dv}{dx} = \frac{v+1}{v-1} - v \Rightarrow x \frac{dv}{dx} = \frac{v+1-(v^2-v)}{v-1}$$

$$\text{or } x \frac{dv}{dx} = \frac{-v^2+2v+1}{v-1} \Rightarrow x \frac{dv}{dx} = -\frac{(v^2-2v-1)}{v-1}$$

$$\Rightarrow \frac{v-1}{v^2-2v-1} dv = -\frac{dx}{x}$$

$$\text{Multiplying by 2, } \frac{2v-2}{v^2-2v-1} dv + 2 \frac{dx}{x} = 0$$

$$\Rightarrow \int \frac{2v-2}{v^2-2v-1} dv + 2 \int \frac{dx}{x} = 0$$

$$\Rightarrow \log(v^2-2v-1) + 2 \log x = \log c$$

$$\Rightarrow \log(v^2-2v-1)x^2 = \log c \Rightarrow (v^2-2v-1)x^2 = c$$

$$\Rightarrow \left(\frac{y^2}{x^2} - 2\frac{y}{x} - 1\right)x^2 = c \Rightarrow y^2 - 2xy - x^2 = c$$

$$(4) \quad \frac{dy}{dx} = \frac{y + \sqrt{x^2 + y^2}}{x} \quad \text{Put } y = vx$$

$$x \frac{dv}{dx} = \sqrt{1+v^2}$$

$$\text{soln } y + \sqrt{x^2 + y^2} = cx^2$$

Linear diff. eq

A diff. eq is said to be linear when the dependent variable and its derivatives occur only in first degree and are not multiplied together.

Eg (1) $x \frac{dy}{dx} + y = 1$ (2) $\frac{dy}{dx} - y = e^{2x}$

The diff. eq $y \frac{dy}{dx} + x = 1$ is not a linear diff. eq

General form of a first order linear diff. eq

is $\frac{dy}{dx} + P(x)y = Q(x)$

Method of finding solution

$e^{\int P(x) dx}$ is called the integrating factor
(I.F). Soln of the above is given by

$$y e^{\int P(x) dx} = \int (e^{\int P(x) dx} \cdot Q(x)) dx$$

$$\text{i.e. } y (I.F) = \int (I.F) Q(x) dx$$

Solve the following diff. eq

(1) $\frac{dy}{dx} - y = e^{2x}$ (2) $(x^2+1) \frac{dy}{dx} + 4xy = x$

(3) $\frac{dy}{dx} + y \tan x = \cos^3 x$ (4) $\frac{dy}{dx} - y \tan x = e^{2x} \sec x$

Solutions

(1) $\frac{dy}{dx} - y = e^{2x}$ $P = -1, Q = e^{2x}$

$$I.F. = e^{\int P dx} = e^{\int (-1) dx} = e^{-x}$$

$$\begin{aligned} \text{soln, } y(e^{-x}) &= \int e^{-x} \cdot e^{2x} \cdot dx \\ &= \int e^x dx \\ &= e^x + c \end{aligned}$$

$$\therefore y = e^x (e^x + c)$$

$$(ii) (x^2+1) \frac{dy}{dx} + 4xy = x$$

Dividing by (x^2+1) , we get

$$\frac{dy}{dx} + \frac{4x}{x^2+1} y = \frac{x}{x^2+1}$$

$$P = \frac{4x}{x^2+1}, \quad Q = \frac{x}{x^2+1}$$

$$\begin{aligned} I.F. &= e^{\int \frac{4x}{x^2+1} dx} = e^{2 \int \frac{2x}{x^2+1} dx} = e^{2 \log(x^2+1)} \\ &= e^{\log(x^2+1)^2} = (x^2+1)^2 \end{aligned}$$

$$\text{Solution: } y(x^2+1)^2 = \int (x^2+1)^2 \cdot \frac{x}{x^2+1} dx$$

$$= \int (x^2+1) x dx$$

$$= \int (x^3 + x) dx$$

$$\therefore y(x^2+1)^2 = \frac{x^4}{4} + \frac{x^2}{2} + C \quad \text{is the soln.}$$

$$(3) \frac{dy}{dx} + y \tan x = \cos^3 x$$

$$P = \tan x, \quad Q = \cos^3 x$$

$$I.F. = e^{\int \tan x dx} = e^{\log \sec x} = \sec x$$

$$\text{Soln: } y(IF) = \int (IF) Q dx$$

$$\therefore y \sec x = \int \sec x \cdot \cos^3 x dx$$

$$= \int \cos^2 x dx$$

$$= \int \frac{1 + \cos 2x}{2} dx$$

$$= \frac{1}{2} \left[x + \frac{\sin 2x}{2} \right] + C$$

$$\therefore y \sec x = \frac{1}{4} [2x + \sin 2x] + C$$

$$(4) P = -\tan x, \quad Q = e^x \cdot \sec x$$

$$I.F. = \cos x$$

$$\text{Soln: } y \cos x = e^x + C$$